

FOREWORD

The story of the world's man-made wonders—of architecture, engineered structures, and sublime works of sculpture—is also the story of mankind. These creations commemorate breathtaking tangible achievements, but they go beyond just the material world because they also reveal mankind's dreams and aspirations. And when the wonders are chosen through time and space—as in this book—the cumulative power is extraordinary.

The subjects raised by the exploration of these wonders cover all aspects of life because architecture, art, and engineering are the epitome of mankind's creative spirit. They touch upon the arcane and subjective aspects of aesthetics but also upon the more objective disciplines of science and physics because structures, by their nature, are often obliged to emulate and echo the stupendous power and laws of nature. This functional—almost elemental—approach is expressed best perhaps by the design of heroic bridges, such as the Akashi Kaikyo Bridge in Japan.

But the story of man-made wonders also includes the more pragmatic cut-and-thrust of politics, the exigencies of economics, and the evolution and refinement of building technologies and techniques. Among the most exciting moments in world architecture come when sudden leaps are made, often through audacious, almost revelatory, experimentation. This happened in Europe in the very early 12th century, when the theories of Gothic construction—embracing the notion of a skeletal frame formed by ribs, piers, and buttresses balanced by pinnacles and utilizing the structural power of the pointed arch—emerged with dramatic speed. This offered revolutionary possibilities to traditional stone construction by imbuing it with added strength realized through finely calculated design.

Even theology plays a key role in this story because so many of mankind's most memorable structures are not to do with structural innovation or the practical issues of providing shelter or security for the living, but with the more poetic challenge of honoring the gods and sustaining the dead. And some—like the Karnak temple complex in Luxor, Egypt—suggest answers to the great riddles that have enthralled humanity across the world and through the ages. Does anything survive death? Can the living commune with the dead? Can the dead intercede for the living? And can there be reincarnation—or even resurrection?

This spiritual quest is usually couched in the most complex of architecture, defined by geometry that simulates the apparent order of the natural world or of Divine creation and by ornament that is both symbolic and emblematic. Aachen Cathedral and the Dome of the Rock in Jerusalem are powerful examples.

Many other themes—tantalizing and inspiring, informative and entertaining—arise when contemplating the wonders in this book. One is the thrilling role of technology—not at least a dry topic. From the early 19th century onward, architecture of the Western world was transformed in majestic manner by the application of new building materials, primarily cast iron and wrought iron and then steel and reinforced concrete, to make unheralded building types possible—notably the skyscraper with a structural frame and heavily glazed curtain walls, including the Shard in London and the Burj Khalifa in Dubai. And then there is the integration of art and architecture and, in certain parts of the world, the reversal of common practice with buildings and volumes excavated rather than constructed—such as the rock-cut churches of Lalibela in Ethiopia—that make architecture, in a sense, giant carved pieces of sculpture.

And it is intriguing to see the principles of sustainable architecture—the use of cheap and available materials, such as unfired clay or mud, that involves little or no environmental damage or pollution, and techniques of natural insulation and ventilation—being applied creatively in traditional vernacular building. The outstanding example in this book is the Great Mosque at Djenné, Mali, built with sun-dried waferlike bricks of mud, dung, and straw covered with render.

Reading this book is like taking a journey through the world—not only the world of the present, but also of the past because the roots of many of the wonders described lie in antiquity. It is also a journey that is increasingly difficult to make in reality as the world becomes more hostile and more volatile, with key sites becoming practically inaccessible. And, in a few cases, the journey is impossible because some of the wonders shown here—notably Palmyra in Syria—have been, in an unthinkable manner, badly damaged and in part destroyed in very recent times. This gives the book another dimension, and a very important one. It not only displays wonders in telling detail that are now hard to see, but also documents wonders that are no more. In this respect, it is a solemn reminder. Never take what we love for granted, and be prepared to fight—if necessary—for our man-made wonders that do much to tell the story of mankind and that make the world such an exciting place.

DAN CRUICKSHANK
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